



Fr. C. Rodrigues Institute of Technology, Vashi
(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

Department of Information Technology
Abstract Major Project

Academic Year	2026-27
Proposed Title of the Project	Agentic AI for Smart Procurement
Key Domain/ Technology:	Manufacturing Industry/Agentic AI\
Complexity & Cost	Complexity-Medium , Cost-Low
Category-I	External- Fractal Analytics
Category-II	Application/Product
Mapping to SDG No. and Names	SDG 8 – Decent Work and Economic Growth SDG 9 – Industry, Innovation and Infrastructure
Mapping to POs	1. Engineering Knowledge 2. Problem Analysis 3. Design/Development of Solutions 4. Conduct Investigations of Complex Problems 5. Engineering Tool Usage 6. The Engineer and The World 7. Ethics 8. Individual and Collaborative Teamwork 9. Communication 10. Project Management and Finance 11. Life-Long Learning
Mapping to PSOs	1. Graduates will be able to apply principles of programming, algorithms, databases, networking, and software engineering to create and develop effective IT solutions for real world problems. 2. Graduates will be able to apply emerging technologies to create innovative IT solutions. 3. Graduates will be able to inculcate self-learning and research attitude



Fr. C. Rodrigues Institute of Technology, Vashi

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

	with collaborative working skills for excelling in continuous personal and professional development
Abstract: Traditional procurement systems in manufacturing industries rely on manual processes and rule-based workflows, leading to delays and inefficiencies. This project proposes an Agentic AI-based Procurement Management System that uses multiple intelligent agents to automate document processing, inventory management, vendor selection, approvals, contract management, and risk analysis. The Proposed system utilizes Apache Kafka, FastAPI, PostgreSQL, Redis, MinIO, Docker, and Kubernetes to improve procurement efficiency, reduce manual effort, and enable faster, more intelligent decision-making.	

Roll No.	Name of the Student	Signature of the Student
1023262	Kaustubh Waje	
5023113	Ashley Fernandes	
5023139	Dikshant Phulpagar	
5023155	Ishwari Shinde	

Name and signature of project guide